Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write Python code to create a dictionary containing mobile names and their prices. Create DataFrame from this dictionary. Display the contents of DataFrame.

[10 Marks]

Q2. Create a package vehicle with a sub package types, sub package type contains modules cars and bikes. Each module should have a function available_models() returning a list of models. Write a python program to display all available models of cars and bikes.

[20 Marks]

OR

Q2. Write a python program to list all files in a current directory. Display total count of .py files and content of any one .py files.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accept a string, Count total number of vowels, consonants and special symbols from it.

[10 Marks]

Q2. Consider a following record in DataFrame IPL.

Player	Team	Category	BidPrice	Runs
Hardik Pandya	Mumbai Indians	Batsman	13	1000
K L Rahul	Kings Eleven	Batsman	12	2400
Andre Russel	Kolkata Knight Riders	Batsman	null	900
Jasprit Bumrah	Mumbai Indians	Bowler	10	200
Virat Kohli	RCB	Batsman	17	3600
Rohit Sharma	Mumbai Indians	Batsman	null	3700

Create above DataFrame in python and write python code for following

a) Retrieve first 2 rows
b) Retrieve last 3 rows
c) Check null values present in DataFrame.
d) Replace null value with mean of that column.
e) Find most expensive player.
f) Print total players per team.

[20 Marks]

OR

Q2. Write a Python program to demonstrate a user-defined exception that checks whether the entered marks are between 0 and 100. Display an appropriate message for invalid input and out-of-range values. Also display Grade of marks.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program that accepts value of m and n. Divides m by n and handles a **ZeroDivisionError** if the user tries to divide by zero.

[10 Marks]

Q2. Write a python program to create a module containing factorial function, power dictionary and list of vowels. Save this as variables.py. Import variables module to print the factorial of n, print the power of n, and second alphabet from the vowels list. (Accept n from user)

[20 Marks]

OR

Q2. Write a python program to accept a file name from user. If file exist display contents of file and number of characters, words and lines in a file, otherwise display appropriate message.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to create a list of elements. Shuffle the elements of a given list and display the list. [Hint: Use random.shuffle()]

[10 Marks]

Q2. Create a following DataFrame named as "data". Write the python code for the following commands.

	Company	Count	Price
Pencil	Apsara	15	250
Pencil	Natraj	20	200
Pen	Cello	25	600
Pen	Parkar	35	900
Eraser	Apsara	20	300

a) Find all rows with the label "Pencil". Extract all columns
b) Change the Eraser count as 25 instead of 20.
c) List only the columns Company and Price.
d) List only rows with labels 'Pencil' and 'Pen'
e) Rename Column Count as Quantity and display DataFrame.

[20 Marks]

OR

Q2. Write a python program for following

a) Create a file which stores the first name, middle name and last name of your 5 friends

b) Display contents of file

c) Append other 5 friend's names

d) Display modified file contents

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accept a string, find all words that do not contain vowels.

[10 Marks]

Q2. Create a package banking with modules account.py and loan.py.

a) account.py should contain function **open_account(name)** that accepts a user's name and prints a message confirming account creation

b) loan.py should contain function **apply_loan(amount)** that accepts a loan amount and prints a confirmation message with the amount

Write a python program to import modules from banking package and demonstrate both functions.

[20 Marks]

OR

Q2. Write a python program to create a list of prime numbers between m and n using **List comprehension**. (Accept the value of m and n from user, m should be less than n)

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program that prompts the user to input two numbers and raises a TypeError exception if the inputs are not numerical.

[10 Marks]

Q2. Create a following DataFrame named as "data". Write the python code for the following

	Company	Count	Price
Pencil	Apsara	15	250
Pencil	Natraj	20	200
Pen	Cello	25	600
Pen	Parkar	35	900
Eraser	Apsara	20	300

a) Find all rows with the label "Pencil". Extract all columns
b) Change the count of Eraser as 25 instead of 20.
c) List only the columns Company and Price.
d) List only rows with labels 'Pencil' and 'Pen'
e) Delete column Count from the above DataFrame and display DataFrame..

[20 Marks]

OR

Q2. Create a Web Page using Flask which contains following information about your Personal Portfolio Page

a) Show your name as a centered heading.
b) List your Qualification details.
c) Add your special skill in scrolling.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accept a string, extract all words that start with a capital letter and remove all special characters from a string. Display modified string. (Accept input from user)

[10 Marks]

Q2. Create a Web Page using Flask which contains following information about your Library Book List

a) Show the library name as a heading.

b) Display all available books in a table with title and author.

c) Add quote about reading in bold and blue color.

[20 Marks]

OR

Q2. Write a python program to accept file name. If file exist copy contents of one file into another file and display contents of new file along with total number of lines, words and characters otherwise give appropriate message.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accepts value of radius from user and calculate the area of a circle using function. (Use 3.14 as the value of π).

[10 Marks]

Q2. Write a python program to create a list, string and tuple in python. Apply random module to display unique multiple random elements from above created sequence type.

[20 Marks]

OR

Q2. Create a Web Page using Flask which contains following information about your college.

a) College Name as a heading in center of web page.

b) Display all the courses available in your college.

c) Image of your College.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accepts a value of n and create a generator that yields even numbers up to n.

[10 Marks]

Q2. Write a python program for following -

a) Create DataFrames Student 1, Student 2 and display contents

b) Join the two DataFrames in third frame with matching records from both sides.

c) Display Third DataFrame.

Student 1

	ID	Name	Age
0	S1	Anita	18
1	S2	Sakshi	23
2	S3	Om	20
3	S4	Raj	21
4	S5	Kanika	20

Student 2

	ID	City	Marks
0	S2	Pune	97
1	S4	Nashik	78
2	S5	Mumbai	86
3	S6	Delhi	66
4	S7	Nanded	54

[20 Marks]

OR

Q2. Write a python program using user define function to accept a number and perform the following task:

a) Check number is Armstrong or not.

b) Calculate and display sum of digits till single digit [e.g. - 558 = 18 = 9]

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to create a DataFrame from list of 10 integer values. Display contents of DataFrame and calculate sum, mean, median, mode, Standard deviation of DataFrame.

[10 Marks]

Q2. Create a package **finance_tools** with modules interest.py and currency.py

a) interest.py should contain function **cal_interest(amount, year)** that accepts amount, year . Calculates and returns simple interest using a fixed rate (for example, 5%).

b) currency.py should contain function **converet(amount)** that accepts amount in INR and converts it into USD. (consider conversion rate 0.012)

Write a python program to import modules from **finance_tools** package and demonstrate both functions.

[20 Marks]

OR

Q2. Write a Python program to create a class that acts as an iterator and generates the factorial series up to n (1! + 2! + 3! + … + n!). (Accept input from the user)

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python code to a create a list of 10 integer numbers. Display square of all elements of list. (Use a lambda function with map())

[10 Marks]

Q2. Create a Flask application that allows users to add and view tasks. The application should have:

a) A form to add a task (task_name).

b) Store tasks in a PostgreSQL database.

c) Display all tasks below the form.

[20 Marks]

OR

Q2. Write a Python program that accepts a string from the user and performs the following tasks:

a) Split the string into individual words.

b) Identify and display all words that do not contain any vowels (a, e, i, o, u), ignore case while checking for vowels.

c) Sort and display these words in alphabetical order**.**

d) Display the total count of such words.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Create a data.csv file containing employee name, designation, salary of an employee. Write a python program to create a DataFrame using data.csv and display the contents of DataFrame with statistical information of dataframe.

[10 Marks]

Q2. Write a Python program that:

a) Retrieves current local time and UTC time.

b) Prints the current time using ctime() and gmtime().

c) Accepts a date-time string input in the format DD-MM-YYYY HH:MM:SS.

d) Parses the string using strptime() and formats it back using strftime() and display it in two custom formats.

[20 Marks]

OR

Q2. Write a python program to create a list, string and tuple in python. Apply random module to display unique multiple random elements from above created sequence type.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1.Write a Python program to get the current time and demonstrate the following functions from the **time** module:

a) Convert the current time in seconds to a readable string

b) Converts time in seconds to a struct_time in UTC

[10 Marks]

Q2. Write a Python program that prompts the user to enter a text file name. If the file exists, open it in read mode, display its contents line by line, and show the current file pointer position after each line using ftell(). Then move the file pointer back to the beginning using seek() and display the file content again. If the file does not exist, display an appropriate error message.

[20 Marks]

OR

Q2. Write a python program for following -

a) Create DataFrames Student 1, Student 2 and display contents

b) Join the two DataFrames in third frame with matching records from both sides.

c) Display Content of Third DataFrame.

Student 1

	ID	Name	Age
0	S1	Anita	18
1	S2	Sakshi	23
2	S3	Om	20
3	S4	Raj	21
4	S5	Kanika	20

Student 2

	ID	City	Marks
0	S2	Pune	97
1	S4	Nashik	78
2	S5	Mumbai	86
3	S6	Delhi	66
4	S7	Nanded	54

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accept a string and substring from user. Check if a particular substring is present at the beginning of the given string, give appropriate message. (Use RE module)

[10 Marks]

Q2. Write a Python program to accept a file name and a byte position as input. If file exist open a file in read mode and move the file pointer to the given byte position, prints the content of the file from that position till end of file otherwise give appropriate message.

[20 Marks]

OR

Q2. Write a python program to create a module containing factorial function, power dictionary and list of vowels. Save this as variables.py. Import variables module to print the factorial of n, print the power of n, and second alphabet from the vowels list. (Accept n from user)

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to get the current time and demonstrate the following functions from the **time** module:

a) Converts time in seconds to a struct_time in UTC

b) Converts a struct_time object into a formatted string.

[10 Marks]

Q2. Create a Web Page using Flask which contains following information about your Library Book List

a) Show the library name as a heading.

b) Display all available books in a table with title and author.

c) Add quote about reading in bold and blue color.

[20 Marks]

OR

Q2. Write a python program to check weather given url is valid or not? (for Ex: https://www.yahoo.com)

Q3 Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program using regular expressions to validate whether a given string matches the pattern of three-letter words such as bit, but, bat, hit, hut, hat.

[10 Marks]

Q2. Consider a following record in DataFrame IPL

Player	Team	Category	BidPrice	Runs
Hardik Pandya	Mumbai Indians	Batsman	13	1000
K L Rahul	Kings Eleven	Batsman	12	2400
Andre Russel	Kolkata Knight Riders	Batsman	7	900
Jasprit Bumrah	Mumbai Indians	Bowler	10	200
Virat Kohli	RCB	Batsman	17	3600
Rohit Sharma	Mumbai Indians	Batsman	15	3700

Create above DataFrame in python and write python code for following

a) Retrieve first 2 rows
b) Retrieve last 3 rows
c) Add null values in DataFrame.
d) Find most expensive player.
e) Print total players per team.

[20 Marks]

OR

Q2. Create a package vehicle with a sub package types, sub package type contains modules cars and bikes. Each module should have a function available_models() returning a list of models. Write a python program to display all available models of cars and bikes.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to accept a string and replace all occurrences of a spaces with a colon and comma with dot. Display modified string.

[10 Marks]

Q2. Write a python program to create a list, string and tuple in python. Apply random module to display unique multiple random elements from above created sequence type.

[20 Marks]

OR

Q2. Write a python program using user define function to accept a number and perform the following task:

a) Check number is Armstrong or not.

b) Calculate and display sum of digits till single digit [e.g. - 558 = 18 = 9]

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to list all files and folders in the current directory using os.listdir().
[10 Marks]

Q2. Write a Python program to validate mobile number based on following criteria

a) The first digit should contain numbers between 6 to 9.
b) The rest 9 digit can contain any number between 0 to 9.
c) The mobile number can have 11 digits also by including 0 at the starting.
d) The mobile number can be of 12 digits also by including 91 at the starting

[20 Marks]

OR

Q2. Build a package education with a sub package courses, containing modules science.py and arts.py. Each module should have a function get_subjects(). Write a python program to display all subjects.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to shuffle a deck of card. Display the cards obtained by a user in 5 attempts.

[10 Marks]

Q2. Write a python program to display all entries in a current directory and identify each as file or folder. [Use getcwd() function to retrieve current directory]

[20 Marks]

OR

Q2. Consider a following record in DataFrame IPL

Player	Team	Category	BidPrice	Runs
Hardik Pandya	Mumbai Indians	Batsman	null	1000
K L Rahul	Kings Eleven	Batsman	12	2400
Andre Russel	Kolkata Knight Riders	Batsman	7	900
Jasprit Bumrah	Mumbai Indians	Bowler	10	null
Virat Kohli	RCB	Batsman	null	3600
Rohit Sharma	Mumbai Indians	Batsman	15	3700

Create above DataFrame in python and write python code for following

a) Retrieve first 2 rows
b) Retrieve last 3 rows
c) Check null values in DataFrame.
d) Replace null values with mean of that column.
e) Find most expensive player.
f) Print total players per team.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to accept a string, Count total number of vowels, consonants and special symbols from it.

[10 Marks]

Q2. Write a Python program to accept a file name and a byte position as input. If file exist open a file in read mode and move the file pointer to the given byte position, prints the content of the file from that position till end of file otherwise give appropriate message.

[20 Marks]

OR

Q2. Create a **Pandas DataFrame** using the following student data with columns and perform the following tasks:

school	class	name	date_of_birth	age	height	weight	address
S1	S001	Asha	15/05/2002	12	173	35	Street1
S2	S002	Fransis	17/05/2002	12	192	32	Street2
S3	S003	Charlie	16/02/1999	13	186	33	Street3
S4	S004	David	25/09/1998	14	167	30	Street4
S5	S005	Shyam	11/05/2002	12	151	31	Street5
S6	S006	Eashel	15/09/1997	16	159	32	Street6

a) Split the DataFrame into groups based on the school column using the GroupBy operation.

b) Check and display the type of the GroupBy object created.

c) Calculate the mean, minimum, and maximum values of the age column for each school group.

d) Count the number of students whose weight is greater than 32.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Create a DataFrame from a dictionary {Subject_Name : Marks} and display the first 5 rows and last 5 rows of DataFrame.

[10 Marks]

Q2. Write a Python program that accepts a filename from the user, opens the file in read mode and display the contents of files. The program should compute and display the total number of characters (including spaces), words, and lines present in the file. If file does not exist give appropriate message.

[20 Marks]

OR

Q2. Create a package banking with modules account.py and loan.py.

a) account.py should contain function **open_account(name)** that accepts a user's name and prints a message confirming account creation.

b) loan.py should contain function **apply_loan(amount)** that accepts a loan amount and prints a confirmation message with the amount.

Write a python program to import modules from banking package and demonstrate both functions.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program that accepts value of m and n. Divides m by n and handles a **ZeroDivisionError** if the user tries to divide by zero.

[10 Marks]

Q2. Write a python program for following -

a) Create DataFrames Student 1, Student 2 and display contents

b) Merge two DataFrames in third frame .

c) Display contents of third DataFrame.

Student 1

	ID	**Name**	**Age**
0	S1	Anita	18
1	S2	Sakshi	23
2	S3	Om	20
3	S4	Raj	21
4	S5	Kanika	20

Student 2

	ID	**City**	**Marks**
0	S2	Pune	97
1	S4	Nashik	78
2	S5	Mumbai	86
3	S6	Delhi	66
4	S7	Nanded	54

[20 Marks]

OR

Q2. Write a Python program to create a class that acts as an iterator and generates the factorial series up to n (1! + 2! + 3! + … + n!). (Accept input from the user)

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program that defines a function productInfo() which accepts a product name and its price. The function should display the product name only if its price is less than Rs.100. If no price is provided, the function should assume the default price to be Rs.100.

[10 Marks]

Q2. Write a Python program that:

a) Retrieves current local time and UTC time.

b) Prints the current time using ctime() and gmtime().

c) Accepts a date-time string input in the format DD-MM-YYYY HH:MM:SS.

d) Parses the string using strptime() and formats it back using strftime() and display it in two custom formats.

[20 Marks]

OR

Q2. Write a Python program that accepts a string from the user and performs the following tasks:

a) Split the string into individual words.

b) Identify and display all words that do not contain any vowels (a, e, i, o, u), ignore case while checking for vowels.

c) Sort and display these words in alphabetical order**.**

d) Display the total count of such words.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to extract number from a given string and display extracted number.

[10 Marks]

Q2. Write a Python program that calculates the pass or fail status of multiple students.

a) Define a function **check_pass_fail(marks)** to return "Pass" if marks ≥ 40, otherwise "Fail".

b) Accept number of students and their marks.

c) Display each student's result.

d) Print the total count of students who passed and failed.

[20 Marks]

OR

Q2. Create a Web Page using Flask which contains following information about your college.

a) College Name as a heading in center of web page.

b) Display all the courses available in your college.

c) Image of your College.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program to shuffle a deck of card. Display the cards obtained by a user in 5 attempts.

[10 Marks]

Q2. Create a package **finance_tools** with modules interest.py and currency.py

a) interest.py should contain function **cal_interest(amount, year)** that accepts amount, year . Calculates and returns simple interest using a fixed rate (for example, 5%).

b) currency.py should contain function **converet(amount)** that accepts amount in INR and converts it into USD. (consider conversion rate 0.012)

Write a python program to import modules from **finance_tools** package and demonstrate both functions.

[20 Marks]

OR

Q2. Write a python program to check weather given url is valid or not?

(for Ex: https://www.yahoo.com)

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a python program to remove all whitespaces from a given string. Display modified string. [10 Marks]

Q2. Create a package vehicle with a sub package types, sub package type contains modules cars and bikes. Each module should have a function available_models() returning a list of models. Write a python program to display all available models of cars and bikes.

[20 Marks]

OR

Q2. Create a following DataFrame named as "data". Write the python code for the following commands.

	Company	Count	Price
Pencil	Apsara	15	250
Pencil	Natraj	20	200
Pen	Cello	25	600
Pen	Parkar	35	900
Eraser	Apsara	20	300

a) Find all rows with the label "Pencil". Extract all columns
b) Change the Eraser count as 25 instead of 20.
c) List only the columns Company and Price.
d) List only rows with labels 'Pencil' and 'Pen'
e) Add a new column named **Colour** to the DataFrame and display DataFrame..

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program that generates a random radius and computes the circumference of a circle using the math module. (Use 3.14 as the value of π)

[10 Marks]

Q2. Write a python program to check the validity of a password given by the user. The password should satisfy the following criteria:

a) Contain at least 1 letter between a and z
b) Contain at least 1 number between 0 and 9
c) Contain at least 1 character from $, #, @
d) Minimum length of password: 6
e) Maximum length of password: 12

[20 Marks]

OR

Q2. Write a python program to list all files in a current directory. Display total count of .py files and content of any one .py files.

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

Duration: 3 Hours **Maximum Marks:35**

Q1. Write a Python program using regular expressions to validate whether a given string matches the pattern of three-letter words such as cat, cot, cut, mat, met, mud.

[10 Marks]

Q2. Build a package education with a sub package courses, containing modules science.py and arts.py. Each module should have a function get_subjects(). Write a python program to display all subjects.

[20 Marks]

OR

Q2. Write a python program to display all entries in a current directory and identify each as file or folder. [Use getcwd() function to retrieve current directory]

Q3. Viva [5 Marks]

Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-271-VSC-P

Advanced Python Programming

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Q1. Write a python program to create a DataFrame using Iris.csv file. Display the contents of DataFrame and statistical information of DataFrame.

[10 Marks]

Q2. Write a Python program that accepts text file name and a string. If file exist, then appends string to its content and display the contents of the file otherwise gives appropriate message.

[20 Marks]

OR

Q2. Write a Python program to validate mobile number based on following criteria

a) The first digit should contain numbers between 6 to 9.

b) The rest 9 digit can contain any number between 0 to 9.

c) The mobile number can have 11 digits also by including 0 at the starting.

d) The mobile number can be of 12 digits also by including 91 at the starting

Q3. Viva [5 Marks]

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Q1. Write a program to replace all occurrences of ‘a’ with * in a String. (Ex. Samidha then output is S*midh*). [10 Marks]

Q2. Write a Python program that calculates the pass or fail status of multiple students.

a) Define a function **check_pass_fail(marks)** to return "Pass" if marks ≥ 40, otherwise "Fail".
b) Accept number of students and their marks.
c) Display each student’s result.
d) Print the total count of students who passed and failed.

[20 Marks]

OR

Q2. Write a Python program that:

a) Retrieves current local time and UTC time.
b) Prints the current time using ctime() and gmtime().
c) Accepts a date-time string input in the format DD-MM-YYYY HH:MM:SS.
d) Parses the string using strptime() and formats it back using strftime() and display it in two custom formats.

Q3. Viva [5 Marks]